

PRODUCT WHITEPAPER

Business Finder

A local discovery platform that helps people find what is open, nearby, and actually happening — and helps local businesses stay visible beyond a static name and star rating.

Product Business Finder (Exchange)

Demo market Toronto, Canada

Audience Customers · Local business owners · Product & engineering stakeholders

Version 1.0 · August 2026

Live demo shope.hunnipages.com

IN ONE SENTENCE

Business Finder is a city-scale discovery app that combines natural-language search, a live open/closed map, and living business profiles so customers can decide in seconds — and owners can publish what is happening today without needing a marketing team.

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1. The problem

Most map and review apps are excellent at answering “where is this place?” They are weaker at answering the questions people actually ask day to day:

- “Where can I get vegan food that is open right now?”
- “Is there a pharmacy nearby that is open late?”
- “What is happening around me today — not last year?”

Listings often freeze a business into a name, address, and star rating. Fresh offers, stock arrivals, events, and short-notice hour changes rarely surface. Owners who are not already famous struggle to stay visible. Customers waste time calling, walking, or scrolling for information that should be obvious.

2. The product idea

Business Finder is built around three capabilities that fill those gaps:

Living profiles

Each business page is a timeline of what is happening today — offers, arrivals, events — not a static brochure.

Plain-language search

People type the way they speak. The app turns that into filters and shows how it understood the question.

Live open / closed map

Green means open now, red means closed, and recent posts ripple on the map so freshness is visible at a glance.

Design principle: help someone decide in under ten seconds — go now, go later, or skip — using open status, distance, freshness, and quality together, not rating alone.

3. Who it is for

Customers

Residents and visitors looking for food, errands, services, and nightlife around them. They want honest open status, nearby options, and a sense of what is new without reading long reviews first.

Business owners

Local shops, cafés, clinics, gyms, and services that want a simple way to post updates, see what drives calls and directions, and compete on freshness — not only on ad spend.

4. What customers get

Natural-language search

Type questions such as “vegan food open now” or “a pharmacy that is open late.” Results appear with a short explanation of how the query was interpreted, so the product feels transparent rather than mysterious.

Map + list discovery

Browse by category or search, then scan a map that mirrors the list. Hovering a result pans the map. Open businesses are easy to spot; places that posted recently are highlighted so “what is alive right now” is visible without opening every profile.

Living business pages

Each profile shows hours with “open now / closing soon / opens in ...”, a feed of updates, reviews, and one-tap Call, Directions, and Save actions. Saves become favourites; followed places surface in a personal feed.

City happening feed

A city-wide stream of posts (filterable by category), plus a “Following” tab with unread badges when signed in. The goal is local pulse, not a social network.

5. What business owners get

- **Fast updates** — post an offer, arrival, event, or news item in roughly ten seconds, with optional expiry and promotion on paid plans.
- **Insights** — 14-day profile views with week-over-week trend, plus counts for directions, calls, and followers.
- **Per-post views** — see which updates actually attract attention.
- **Plan signal** — free / featured / premium placement is modelled so visibility can become a clear revenue lever later.

Demo account	Email	Password	What you see
Customer	customer@demo.test	demo1234	Favourites and follow notifications
Business owner	owner@demo.test	demo1234	Dashboard for four businesses: post updates and view insights

6. How search works

Search uses one interface and two interpreters. The product always tries to answer; AI is an upgrade, not a dependency.

Interpreter	When it runs	Strength
Built-in lexicon (default)	Always available — no API key, no network	Instant mapping of phrases to filters: open now, late night, price, quality, proximity, freshness, cuisine, diet, category

Interpreter	When it runs	Strength
Claude (optional)	When <code>ANTHROPIC_API_KEY</code> is set	Handles unusual phrasing via a JSON schema; any failure falls back to the built-in interpreter

Reliability rule: search never fails because of an AI outage. The UI is told which interpreter answered, so demos stay honest.

7. How open / closed is decided

Hours are stored as minutes after local midnight for each business's own timezone offset. Closing times past midnight are allowed (for example, 20:00-02:00 becomes 1200 → 1560), so late-night venues report correctly after midnight.

Beyond a binary open/closed flag, the engine surfaces decision-useful detail: closing in 20 min or opens in 40 min. That is often what changes a customer's plan.

The Toronto demo uses Eastern Daylight Time (UTC−4) so status matches local Canadian hours during the summer season.

8. How results are ranked

Results are scored like a good local guide, not a raw database dump. Five signals are blended:

1. **Proximity** — closer places rise when location is known.
2. **Open now** — open businesses are preferred when that intent matters.
3. **Freshness** — recent posts boost visibility.
4. **Rating** — quality still counts, but not alone.
5. **Subscription plan** — featured and premium placement is the modelled revenue lever.

9. Technology overview

The stack is intentionally small so the product can run with zero external database setup and stay easy to demo or deploy.

Layer	Choice	Why it fits
Web app	React + Vite	Fast UI, map/list discovery, light & dark themes, mobile tab bar
API	Node.js + Express	Simple REST surface for search, profiles, auth, owner tools
Data	Built-in SQLite (<code>node:sqlite</code>)	No native deps, seeds on first boot, ideal for demos and early markets
Auth	scrypt passwords + HMAC session tokens	Enough for demo accounts; production should move to httpOnly cookies
Maps	Leaflet / React-Leaflet	Open map tiles with open/closed pin styling

Deployment shape (this environment)

- Frontend and API run under **PM2**.
- **Nginx** serves the public host and proxies `/api` to the backend so the browser stays on one origin.
- Demo city centre: **Toronto** (configurable via seed environment variables for any city).

10. Demo guide

Suggested five-minute walkthrough

1. Open Discover and read the city label — confirm Toronto.
2. Search: “coffee open now” or “pharmacy open late”. Notice the interpretation chip above results.
3. Toggle the map: green / red pins, ripple on recently posted places.
4. Open a business profile: hours status, update timeline, Call / Directions.
5. Visit Happening for the city feed; sign in as the customer to save favourites.
6. Sign in as the owner to post an update and open Insights.

What is pre-loaded

On first boot the API seeds about **30 businesses**, roughly **50 live updates**, and two weeks of interaction history. The demo is never empty and needs no manual database setup.

11. Roadmap notes

The current build is a complete, working product suitable for demos and early pilots. A few areas are intentionally demo-grade and ready for a production hardening pass:

- **Sessions** — move from localStorage tokens to httpOnly cookies with refresh rotation.
- **Scale** — SQLite is ideal early; multi-city scale benefits from Postgres + PostGIS for proximity queries.
- **Media** — photos currently use placeholders; real listings need uploads and a CDN.
- **Payments** — featured placement and promoted posts are modelled; a payment provider is the next commercial step.
- **Push** — notifications are in-app today; push needs a service worker and push service.

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This document describes the product as implemented in the Exchange / Business Finder codebase.